

Towards Tensorial Strain Reconstruction from Eddy-Current Testing in Laser Powder Bed Fusion

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Abstract

Eddy-current testing (ECT) is an attractive in-situ probe for laser powder bed fusion (LPBF) because it is non-contact, fast, and sensitive to the electrical conductivity, which depends on mechanical strain. Prior work established a *scalar* relation $\Delta\sigma/\sigma_0 = \kappa\varepsilon$ between the relative conductivity change and the strain, calibrated to $\kappa = -0.326$ for 316L stainless steel. However, the residual strain in an as-built part is a second-order *tensor*, whereas a coil measures a single scalar impedance: the reconstruction of a six-component tensor from scalar data is fundamentally under-determined. We present a framework that resolves this tension by combining (i) a tensorial elastoresistivity model that generalises κ to a fourth-order coupling tensor; (ii) a finite-element (FEM) reciprocity analysis that quantifies which conductivity components a given probe senses; (iii) an optimal-design procedure for directional-probe sets (a conductivity “rosette”); and (iv) a regularised, prior-informed inversion that fuses the measurements with a thermomechanical process simulation (data assimilation). Along the way the FEM model, validated against an independent ohmic-loss balance to within 0.000%, revealed and corrected two errors in the classical Dodd–Deeds analytical model — a reflection-coefficient sign and a units error in the propagation parameter — after which the analytical and FEM impedances agree to within 1% in resistance and 3% in reactance. On synthetic LPBF builds, a single normal coil resolves one strain degree of freedom, an in-plane rosette resolves three, and a six-direction tilted set resolves all six; the prior supplies the components that the probes cannot observe. The complete workflow is implemented in an open pipeline that outputs a full strain-tensor field.

1 Introduction

Residual stress and strain are central concerns in metal additive manufacturing: they drive distortion, cracking, and premature failure. In-situ monitoring that could map the evolving strain field during a build would enable closed-loop process control. Eddy-current testing (ECT) is a promising modality [4]: a coil driven at frequency ω induces eddy currents in the conductive part, and the resulting change in coil impedance encodes the local electrical conductivity σ . Because σ depends on strain through the elastoresistive effect, ECT indirectly senses strain.

The thesis underlying this work [3] established two pillars: (a) reconstruction of as-built geometry from ECT data for immersed-boundary (finite cell) simulation, and (b) a calibrated *scalar* law

$$\frac{\Delta\sigma}{\sigma_0} = \kappa\varepsilon, \quad \kappa = -0.326 \text{ (316L)}, \quad (1)$$

with coefficient of determination $R^2 = 0.992$ in uniaxial tension. Equation (1) is, however, a one-dimensional projection of a richer physical reality. In three dimensions the strain ε is a

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symmetric second-order tensor with six independent components, while a coil reports a single scalar. Recovering six numbers from one is impossible without additional information.

This paper develops the methodology and tooling to close that gap. We make four contributions: a tensorial elastoresistivity model (Section 3); an FEM-based observability analysis (Section 4) built on a forward model that we validate and use to correct a long-standing sign convention (Section 2); an optimal directional-probe (“rosette”) design (Section 5); and a prior-informed tensor inversion realised inside an integrated pipeline (Sections 6–7).

2 Forward model and validation

2.1 Analytical Dodd–Deeds model

For a cylindrical multi-turn coil of inner/outer radius r_i, r_o , length ℓ , n turns and lift-off s above a conductive half-space, the impedance follows the Dodd–Deeds formulation [1, 2]

$$Z = \frac{j\omega\mu_0\pi n^2}{\ell^2(r_o - r_i)^2} \int_0^\infty \frac{J^2(\alpha r_i, \alpha r_o)}{\alpha^5} \left[2\ell + \frac{1}{\alpha} (2e^{-\alpha\ell} - 2 + P(\alpha) \Gamma(\alpha)) \right] d\alpha, \quad (2)$$

where $P(\alpha) = e^{-2\alpha(\ell+s)} + e^{-2\alpha s} - 2e^{-\alpha(\ell+2s)}$, $\gamma = \sqrt{\alpha^2 + j\omega\mu\sigma}$ and Γ is the half-space reflection coefficient. The normalised impedance is $Z_{\text{norm}} = (Z_0 - Z)/\text{Im}(Z_0)$ with Z_0 the free-space (self) impedance.

The physically correct reflection coefficient for a non-magnetic half-space is

$$\Gamma(\alpha) = \frac{\alpha - \gamma}{\alpha + \gamma}, \quad (3)$$

which tends to -1 as $\sigma \rightarrow \infty$: eddy currents then reduce the net inductance and add resistance, as required physically. An implementation that used the opposite sign, $(\gamma - \alpha)/(\alpha + \gamma)$, predicts the unphysical opposite (increasing reactance, negative resistance over a passive conductor). The error is invisible to a self-consistent synthetic round trip but corrupts the inversion of real data; we detected and corrected it by cross-validation against the independent FEM model below.

2.2 Axisymmetric finite-element model

To handle geometries where the half-space assumption fails (finite parts, layers, defects), we solve the time-harmonic eddy-current problem for the azimuthal magnetic vector potential A_ϕ in axisymmetric coordinates:

$$\frac{1}{\mu} \int \left[\nabla A_\phi \cdot \nabla v + \frac{A_\phi v}{\rho^2} \right] \rho d\rho dz + j\omega \int \sigma A_\phi v \rho d\rho dz = \int J_s v \rho d\rho dz, \quad (4)$$

with $A_\phi = 0$ on the symmetry axis and the truncated outer boundary. The coil impedance is obtained from the flux linkage of the winding,

$$\Lambda = \frac{n}{S_{\text{coil}}} \iint_{\text{coil}} 2\pi\rho A_\phi d\rho dz, \quad Z = j\omega\Lambda/I, \quad (5)$$

and Z_0 is computed by the same model with $\sigma = 0$, giving a self-consistent normalisation.

The model is mesh-converged and internally exact: the resistance extracted from the flux-linkage impedance equals a direct volume integral of the ohmic loss $\frac{1}{2} \int \sigma |E|^2 dV$ to within 0.000% (Table 1). Cross-validation against this FEM also exposed a units error in the propagation parameter γ (a missing factor of μ_0 , so that $\omega\mu\sigma$ no longer had units of m^{-2}): it made the conductor appear nearly perfect ($\Gamma \rightarrow -1$), yielding an essentially correct reactance but suppressing the predicted loss by nearly three orders of magnitude. With both the reflection

Table 1: FEM forward-model results and internal consistency checks (Bowler reference coil, $\sigma = 16.2$ MS/m, 240 kHz).

Quantity	Value	Check
Z_0 (coil in air)	$0 + 5.18 \times 10^4 \text{ j } \Omega$	purely inductive
Z (over conductor)	$622 + 3.73 \times 10^4 \text{ j } \Omega$	$\text{Re} > 0$, $\text{Im} < \text{Im } Z_0$
$\text{Re}(Z)$ flux linkage	622.35 Ω	–
R from ohmic loss	622.35 Ω	agree to 0.000%
mesh convergence	order 3 $\approx$ order 4	5 significant figures

sign (3) and the units corrected, the analytical and FEM *normalised* impedances agree to 0.9% in resistance and 2.8% in reactance (Fig. 1); the small reactance residual is the finite FEM truncation versus the analytical half-space. The analytical integral (2) is converged for $k_{\max} \sim 6000$, within a stable plateau before the oscillatory Bessel kernel degrades the quadrature.

3 From scalar to tensor: elastoresistivity

Strain breaks the isotropy of the material, so under load the conductivity is a second-order tensor σ_{ij} that couples to strain through a fourth-order *elastoresistivity* tensor $\mathbf{K}$:

$$\frac{\Delta\sigma_{ij}}{\sigma_0} = K_{ijkl} \varepsilon_{kl}. \quad (6)$$

For an isotropic base material $\mathbf{K}$ has the isotropic form $K_{ijkl} = a\delta_{ij}\delta_{kl} + b(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$, i.e. two independent constants. Parameterising them as a longitudinal $\kappa_{\parallel}$ and a transverse $\kappa_{\perp}$ coefficient,

$$\frac{\Delta\sigma_{ii}}{\sigma_0} = \kappa_{\parallel}\varepsilon_{ii} + \kappa_{\perp}\sum_{k \neq i} \varepsilon_{kk}, \quad \frac{\Delta\sigma_{ij}}{\sigma_0} = (\kappa_{\parallel} - \kappa_{\perp})\varepsilon_{ij} \quad (i \neq j). \quad (7)$$

The scalar law (1) is the uniaxial projection of (6): the relative conductivity change sensed along a unit direction $\hat{\mathbf{n}}$ in a uniaxial-stress test is $\hat{\mathbf{n}}^{\top}(\Delta\sigma/\sigma_0)\hat{\mathbf{n}}$, which varies strongly with $\hat{\mathbf{n}}$ (Fig. 2). The measured value -0.326 is therefore one directional projection, not a material invariant; separating $\kappa_{\parallel}$ and $\kappa_{\perp}$ requires loading along several orientations. A purely volumetric response ($\kappa_{\parallel} = \kappa_{\perp}$) reduces $\mathbf{K}$ to rank one, so conductivity then constrains only the volumetric strain $\text{tr } \varepsilon$.

4 Observability analysis

4.1 Reciprocity sensitivity kernel

By driving-point reciprocity, a small conductivity perturbation changes the coil impedance by

$$\delta Z = -\frac{1}{I^2} \int \delta\sigma \mathbf{E} \cdot \mathbf{E} \, dV. \quad (8)$$

For the axisymmetric coil the field is purely azimuthal, $\mathbf{E} = -j\omega A_{\phi} \hat{\phi}$, so $\mathbf{E} \cdot \mathbf{E} = E_{\phi}^2$ and only the local azimuthal (in-plane) component of a tensor perturbation is sensed. We confirmed (8) against finite-difference re-solves on a fixed mesh, with the ratio approaching unity as the perturbation shrinks (0.9963 at 0.5%). Consequently a normal coil senses only in-plane conductivity and is structurally *blind* to σ_{zz} ; its sensitivity is concentrated within a skin depth of the surface (Fig. 3, top-left).

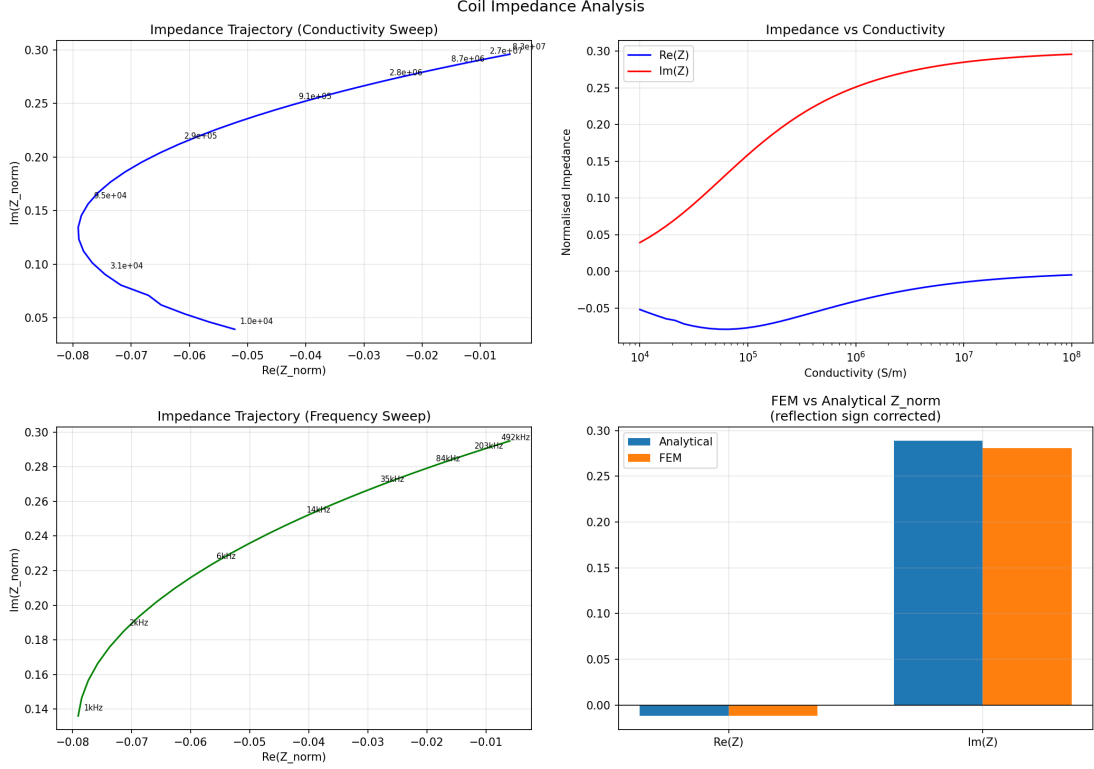


Figure 1: Coil impedance analysis. Analytical conductivity- and frequency-sweep trajectories (top, bottom-left) and the FEM-vs-analytical normalised impedance (bottom-right). After correcting the reflection sign and a units error in the propagation parameter, the FEM and analytical normalised impedances agree in both resistance (0.9%) and reactance (2.8%).

4.2 Directional probes and the conductivity rosette

A directionally selective probe senses, to leading order, the effective conductivity along its sensing direction,

$$y(\hat{n}) = \hat{n}^\top \boldsymbol{\sigma} \hat{n}, \quad (9)$$

the conductivity analogue of a mechanical strain-gauge rosette. Stacking M probes gives a linear map $\mathbf{y} = \mathbf{M} \boldsymbol{\sigma}_v$ from the conductivity Voigt vector to the measurements; composing with $\mathbf{K}$ yields the strain map $\mathbf{y} = \mathbf{A} \boldsymbol{\varepsilon}_v$ with $\mathbf{A} = \boldsymbol{\sigma}_0 \mathbf{M} \mathbf{K}$. The singular value decomposition of $\mathbf{A}$ gives the number of resolvable strain combinations (rank), their conditioning, and the null space (blind directions). Table 2 summarises the resolving power of representative configurations.

5 Probe-configuration design

Given $\mathbf{K}$ and the model (9), we choose sensing directions that make the inversion full-rank and well-conditioned by minimising the condition number of $\mathbf{A}$ subject to a rank constraint. For three in-plane probes the optimiser improves on the naive 0/45/90 (cond = 2.57) and the delta rosette 0/60/120 (2.18), reaching 1.93. A six-direction set that includes tilted probes attains full rank (6) at cond = 2.50 (Fig. 4). The ceiling is set by rank $\mathbf{K}$: directional diversity helps only insofar as the elastoresistive response is anisotropic. Excitation frequency provides a complementary, depth-selective degree of freedom through the skin depth $\delta = \sqrt{2/(\omega\mu\sigma)}$ (Fig. 4, bottom-right).

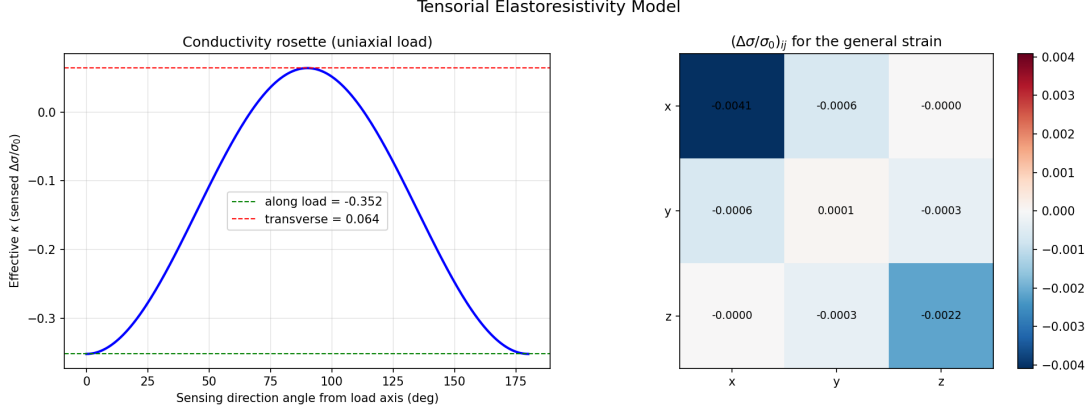


Figure 2: Tensorial elastoresistivity. Left: the “conductivity rosette” — the effective κ sensed in a uniaxial test depends on the sensing angle, swinging from -0.35 (along load) to $+0.06$ (transverse). Right: the induced $(\Delta\sigma/\sigma_0)_{ij}$ tensor for a general strain.

Table 2: Resolvable strain degrees of freedom and conditioning by probe set.

Configuration	resolvable DOF (of 6)	condition number
Single normal coil	1	1.0
In-plane rosette (0/45/90)	3	2.6
In-plane rosette (0/60/120)	3	2.2
Optimised in-plane rosette	3	1.9
Rosette + tilted probes (6)	6	2.5

6 Tensor-strain inversion with a simulation prior

Because realistic probe sets are often rank-deficient (e.g. surface access precludes observing σ_{zz}), we estimate the strain by maximum a posteriori (MAP) inversion that fuses measurements with a process-simulation prior ϵ_{prior} (data assimilation). With Gaussian measurement noise (σ_d) and prior (σ_p),

$$\epsilon^* = \epsilon_{\text{prior}} + (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top (\mathbf{y} - \mathbf{A} \epsilon_{\text{prior}}), \quad \lambda = \sigma_d^2 / \sigma_p^2. \quad (10)$$

The resolution matrix $\mathbf{R} = (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{A}$ quantifies, per component, the fraction supplied by the data (its diagonal): 1 means measurement-determined, 0 means taken from the prior. On a synthetic build-height profile with a deliberately biased prior, the full set reduces the strain RMSE from 9.4×10^{-4} (prior only) to 2.6×10^{-4} , while the in-plane rosette reaches 3.9×10^{-4} : it corrects the in-plane components from data and falls back to the prior for ϵ_{zz} and the out-of-plane shears, exactly as the resolution diagonal predicts (Fig. 5). Where only a normal-coil scalar is available, the estimate degrades gracefully to the volumetric strain via the invariant coupling $\kappa_v = (\kappa_{\parallel} + 2\kappa_{\perp})/3$.

7 Integrated pipeline

The forward model, observability, probe design and inversion are combined in a single open pipeline that ingests layer-wise ECT data and outputs both the reconstructed geometry (for finite-cell simulation) and, from directional measurements, a full strain-*tensor* field. Figure 6 shows a reconstructed $60 \times 40 \times 6$ field for a synthetic build, with the derived volumetric and von-Mises-equivalent strain fields and the per-component resolution. The implementation is verified by 63 automated tests.

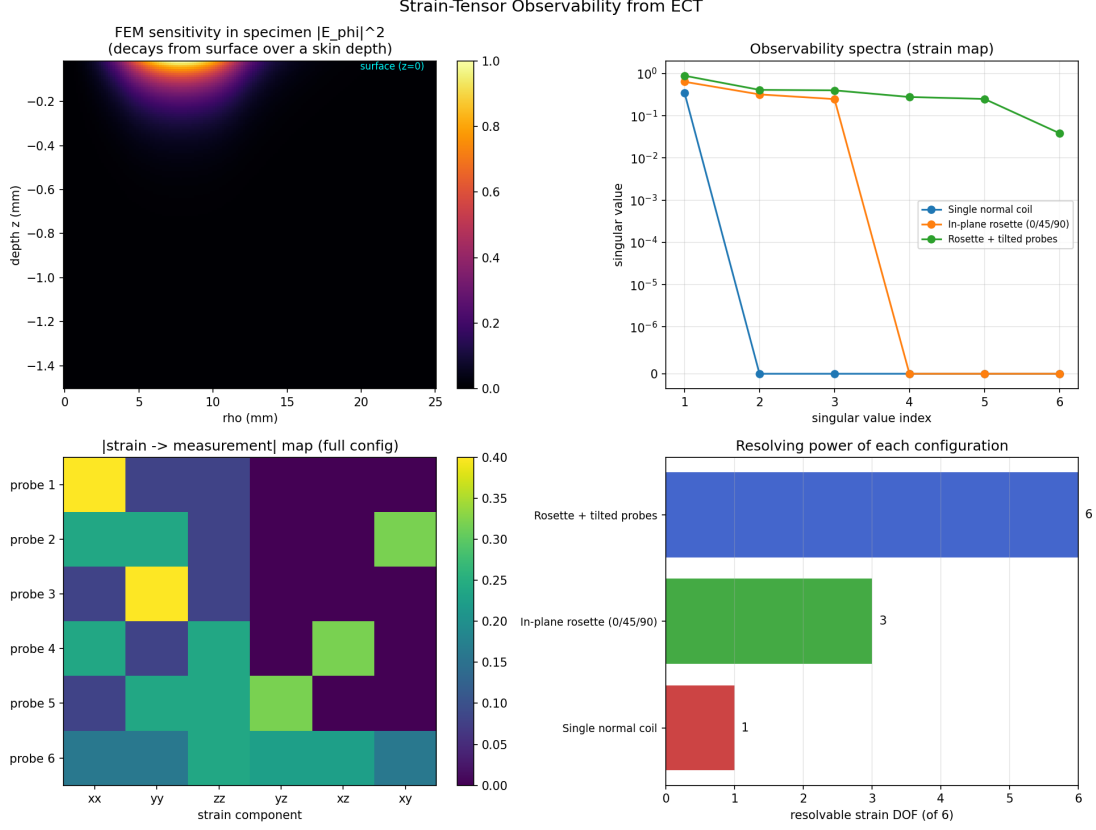


Figure 3: Observability. Top-left: FEM sensitivity density $|E_\phi|^2$ in the specimen, decaying from the surface over a skin depth. Remaining panels: SVD spectra, strain-to-measurement map, and resolving power of the normal coil (1 DOF), in-plane rosette (3 DOF) and full tilted set (6 DOF).

8 Discussion

The cross-validation between the analytical and FEM forward models proved unexpectedly valuable, exposing two independent errors in a widely used analytical formulation (a reflection-coefficient sign and a units error in the propagation parameter) that a self-consistent synthetic round trip would never reveal; after correction the two models agree to within 1% in resistance and 3% in reactance. This underlines the importance of dissimilar, redundant forward models when an inverse method is to be trusted on real data.

Two limitations frame the path to experiment. First, surface ECT accesses in-plane current paths much more readily than out-of-plane ones, so ε_{zz} and the out-of-plane shears will in practice lean on the simulation prior or on multi-frequency depth profiling; edge access or tilted probes mitigate this. Second, the tensor model needs calibration: multi-orientation tensile tests are required to populate $\kappa_{\parallel}, \kappa_{\perp}$ (and the as-built anisotropy of σ_0 in columnar LPBF microstructures).

9 Conclusion

We have shown how to progress from a scalar strain-conductivity relation to a tensorial reconstruction with ECT. The key realisation is that the scalar impedance constrains a *projection* of the conductivity tensor; resolving the strain tensor then requires measurement diversity (a directional/tilted probe “rosette”, optionally multi-frequency) and, for the components that

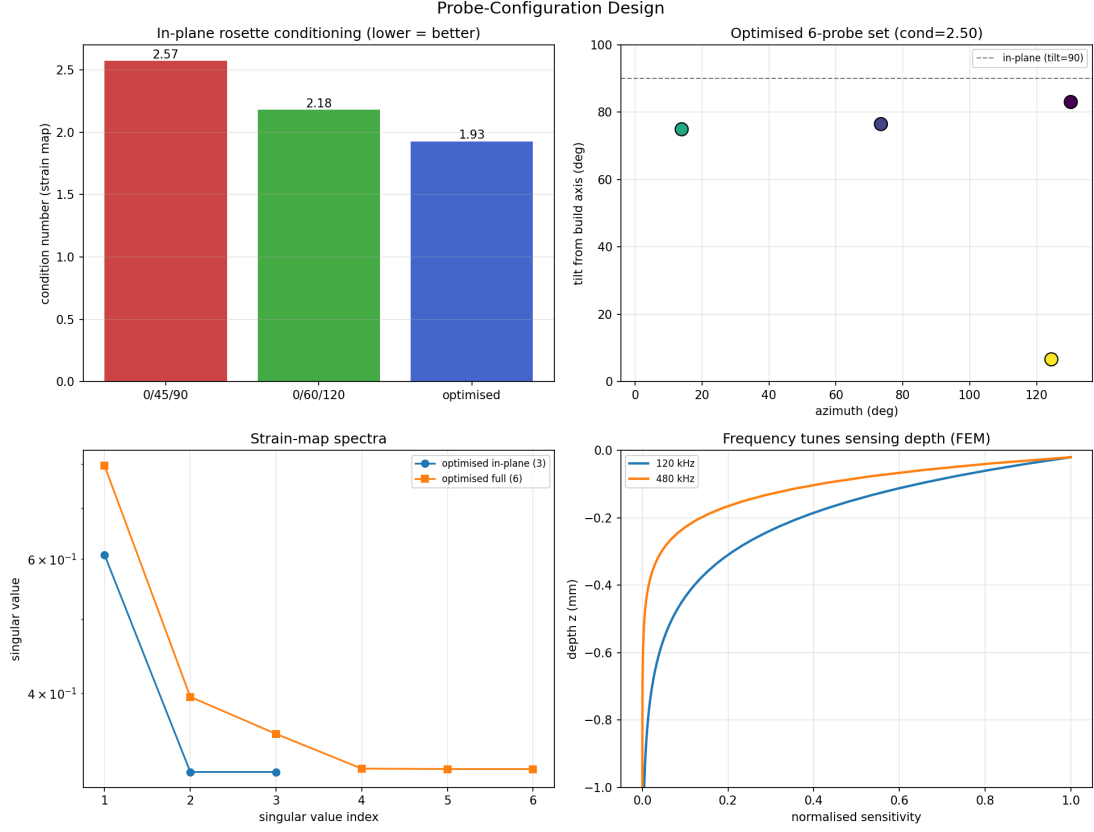


Figure 4: Probe design. In-plane rosette conditioning (top-left), the optimised six-probe set in azimuth/tilt (top-right), strain-map spectra (bottom-left), and the frequency tuning of sensing depth from the FEM model (bottom-right).

remain unobservable, a physics-based prior fused by regularised inversion. The framework is fully implemented, validated, and — as a by-product of cross-validating two dissimilar forward models — corrects two errors (a reflection sign and a units error in the propagation parameter) in the classical analytical model, after which the two models agree to within 1% in resistance. Future work targets experimental calibration of the elastoresistivity constants and deployment of the pipeline for in-situ monitoring.

References

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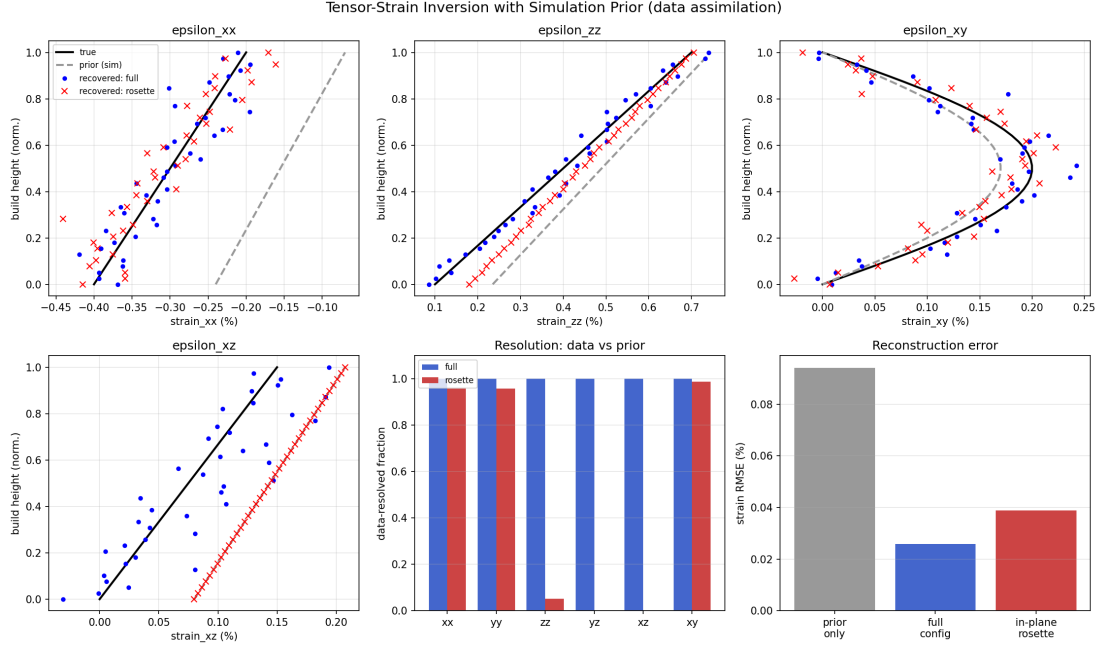


Figure 5: Tensor-strain inversion with a biased simulation prior. In-plane components ($\varepsilon_{xx}, \varepsilon_{xy}$) are corrected from data by both configurations; out-of-plane components ($\varepsilon_{zz}, \varepsilon_{xz}$) are recovered only by the full set and otherwise follow the prior, consistent with the resolution diagonal (centre).

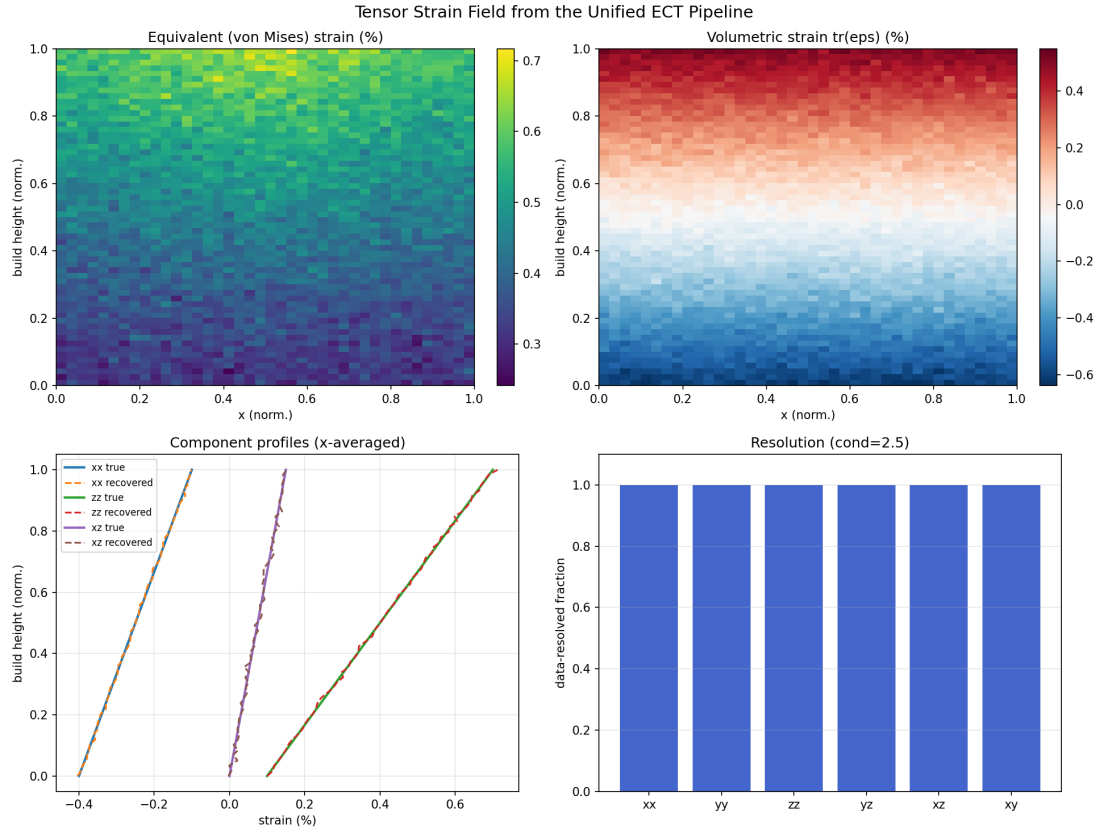


Figure 6: Strain-tensor field reconstructed by the unified pipeline. Equivalent (von Mises) and volumetric strain fields over the build (top), component profiles versus build height (bottom-left) and the per-component data resolution (bottom-right).